# LOOPS.JS

```
// 1. Print numbers 1 to 10

// Write a program using a for loop to print numbers from 1 to 10.

// for (let i = 1; i <= 10; i++) {

//     console.log(i)

// }


// 2. Sum of first N numbers

// Use a while loop to calculate the sum of the first N natural numbers.

// let number = 2;

// let i = 1;

// let sum = 0;

// while (i <= number) {

//     sum = sum + i

//     i = i + 1

// }

// console.log(sum)


// 3. Multiplication table

// Print the multiplication table of a given number using a for loop.

// let res = 16;

// for (let i = 0; i <= 10; i++) {

//     console.log(res * i)

// }
```

```
// 4. Factorial calculation
// Write a program using a while loop to find the factorial of a given number.
// let number = 9;
// let factorial = 2;
// let i = 1;
// while (i <= number) {
//     factorial = factorial * i
//     i = i + 1
// }
// console.log(factorial)

// 5. Reverse counting
// Print numbers from 10 down to 1 using a for loop.
// for (let i = 10; i >= 1; i--) {
//     console.log(i);
// }

// 6. Even numbers up to N
// Use a do-while loop to print all even numbers up to N.
// let i = 1;
// let N = 10;

// do {
//     if (i % 2 == 0) {
//         console.log(i);
//     }
```

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//     i++;
// } while ( i <= N );

// 7. Sum of digits
// Write a program using a while loop to calculate the sum of digits of a given number.
// let number = 123 ;
// let sum = 0;
// while (number > 0) {
//     last= number % 10;
//     sum = sum + last;
//     number= Math.floor(number / 10)
// }
// console.log(sum)

// 8. Fibonacci series
// Generate the first 10 terms of the Fibonacci series using a for loop.
// let a = 0;
// let b = 1;

// for (let i = 1; i <= 10; i++) {
//     console.log(a);

//     let next = a + b;
//     a = b;
//     b = next;
```

```
// }

// 9. Guessing game
// Use a do-while loop to keep asking the user for a number until they guess the correct one.
// let correctNumber = 7;
// let guess = 5;

// do {
//     console.log("Wrong! Try again.");

//     guess++;
// } while (guess !== correctNumber);

// console.log("Correct!");

// 10. Prime number check
// Write a program using a for loop to check if a given number is prime.
// let number = 17;
// let isPrime = true;

// if (number <= 1) {
//     isPrime = false;
// }

// for (let i = 2; i < number; i++) {
//     if (number % i === 0) {
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//         isPrime = false;
//         break;
//     }
// }

// if (isPrime) {
//     console.log("Prime number");
// } else {
//     console.log("Not a prime number");
// }
```

# CONDITIONAL STATEMENT.JS

```
// 1. Write a program that checks if a number is positive, negative, or zero.
// let number= 0;
// if (number == 0) {
//     console.log(`${number} is zero`)
// }
// else if (number> 0) {
//     console.log(`${number} is positive`)
// }
// else{
//     console.log(`${number} is negative`)
// }
```

```
// 2. Even or odd check
// Using an if-else statement, determine whether a given integer is even or odd.
// let number= "a";
// if ( number % 2 === 0) {
//     console.log("Number is even")
// }
// else if ( number % 2 == 1){
//     console.log("number is odd")
// }
// else{
//     console.log("number is zero")
// }

// 3. Largest of two numbers
// Write a program that takes two numbers and prints the larger one using conditional statements.
// let number1 = 220;
// let number2 = 54;
// if (number1 > number2) {
//     console.log(`${number1} is greater`)
// }
// else{
//     console.log(`${number2} is greater`)
// }

// 4. Grade evaluation
```

```
// Using if-else-if, assign grades (A, B, C, D, F) based on a student’s percentage score.
// let marks = 22;
// if (marks > 80) {
//     console.log("Student got A grade")
// }
// else if (marks > 70) {
//     console.log("Student got B grade")
// }

// else if (marks > 60) {
//     console.log("Student got c grade")
// }

// else{
//     console.log("Student got F grade")
// }
// 5. Leap year check
// Write a program that checks if a given year is a leap year using conditional statements.
// let year = 2024;
// if (year % 4 === 0) {
//     console.log("leap year")
// }
// else{
//     console.log("no leap year")
// }
```

```
// 6. Day of week switch

// Use a switch-case to print the name of the day when given a number (1 = Monday, 2 = Tuesday, … 7 = Sunday).

// let number = 3;

// switch (number) {

//     case 1:

//         console.log("Monday")

//         break;

//     case 2:

//         console.log("Tuesday")

//         break;

//     case 3:

//         console.log("Wednesday")

//         break;

//     case 4:

//         console.log("Thursday")

//         break;

//     case 5:

//         console.log("Friday")

//         break;

//     case 6:

//         console.log("Saturday")

//         break;

//     case 7:

//         console.log("Sunday")

//         break;
```

```
//     default:
//         console.log("No day")
//         break;
// }

// 7. Calculator switch
// Create a simple calculator using switch-case that performs addition, subtraction, multiplication, or division based on user input.
// let number1 = 22;
// let number2 = 33;
// let operator = "-";
// switch (operator) {
//     case "+":
//         console.log(number1 + number2)
//         break;
//     case "-":
//         console.log(number1 - number2)
//         break;
//     case "*":
//         console.log(number1 * number2)
//         break;
//     case "/":
//         console.log(number1 / number2)
//         break;
//     default:
```

```
//       break;
// }


// 8. Vowel or consonant
// Write a program that checks whether a given character is a vowel or consonant using switch-case.
// let letter = "a"
// switch (letter) {
//    case "a":
//       console.log("Leter is vowel")
//       break;
//    case "e":
//       console.log("Leter is vowel")
//       break;
//    case "i":
//       console.log("Leter is vowel")
//       break;
//    case "o":
//       console.log("Leter is vowel")
//       break;
//    case "u":
//       console.log("Leter is vowel")
//       break;
//    default:
//       console.log("Letter is consonant")
//       break;
```

```
// }


// 9. Traffic light system
// Using switch-case, print instructions based on traffic light color (Red = Stop, Yellow = Wait, Green = Go).
// let signal = "green";
// switch(signal) {
//     case "red":
//         console.log("Stop the car")
//         break;
//     case "Yellow":
//         console.log("Wait for your turn")
//         break;
//     case "green":
//         console.log("You can go now")
//         break;
//     default:
//         console.log("Stop")
//         break;
// }


// 10. Menu-driven program
// Write a program using switch-case where the user selects from a menu (e.g., 1 = Check Balance, 2 = Deposit, 3 = Withdraw, 4 = Exit).
// let menu = 4;
// switch (menu) {
//     case 1:
```

```
//       console.log("You have selected check Balance")
//       break;
//    case 2:
//       console.log("You have selected Deposit")
//       break;
//    case 3:
//       console.log("You have selected Withdraw")
//       break;
//    case 4:
//       console.log("Exit, Thanks for coming")
//       break;
//    default:
//       console.log("Invalid")
//       break;
// }
```